

Cell Biology and Immunology: Transplantation & Graft vs Host disease

Integrated Biomedical and Clinical Sciences OMFS 513

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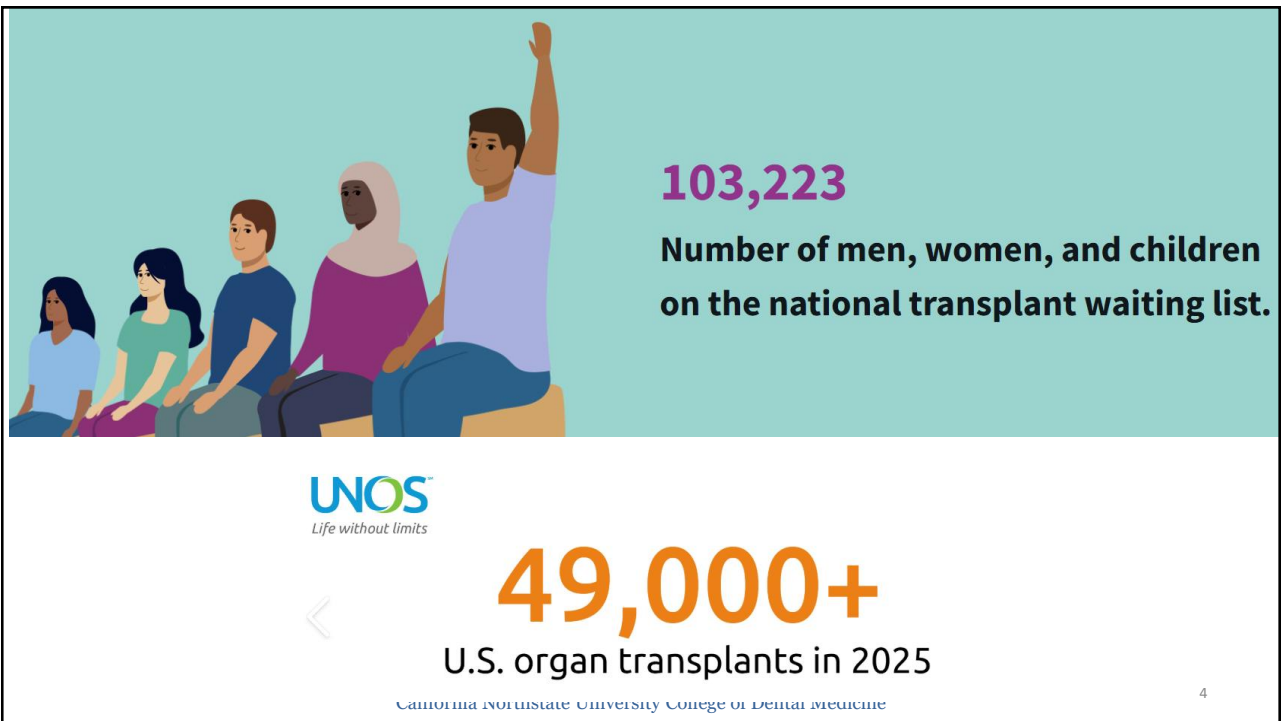
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Learning Objectives

- 1- Compare different types of grafts
- 2- Describe the mechanism of allograft rejection.
- 3- Define graft versus host disease and identify acute and chronic symptoms.
- 4- Describe prevention strategies to graft rejection.

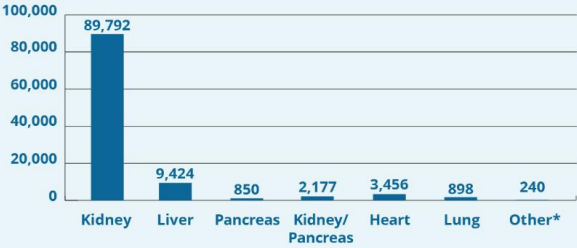
Outline

1. Histocompatibility
2. Types of grafts.
3. Transplant outcomes
 - I. Transplant rejection.
 - II. Graft versus host disease.
4. Prevention of allograft rejection.



Patients on the Waiting List by Organ

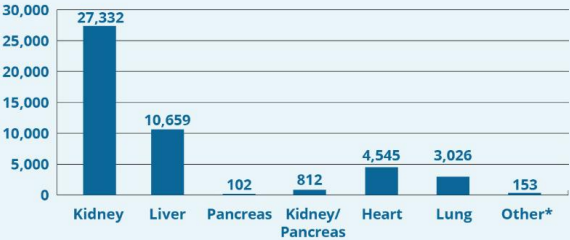
As of September 2024



*Other includes allograft transplants like face, hands, and abdominal wall.
Based on OPTN data as of September 15, 2024. Data subject to change based on future data submission or correction. Totals may be less than the sums due to patients included in multiple categories.

Transplants Performed by Organ

In 2023



*Other includes allograft transplants like face, hands, and abdominal wall.
Based on OPTN data as of September 15, 2024. Data subject to change based on future data submission or correction. Totals may be less than the sums due to patients included in multiple categories.

**Calculated Panel of Reactive Antibody (CPRA)
is matching system to determine the % of
reactive antibodies leading to a possible
transplant rejection**

Histocompatibility

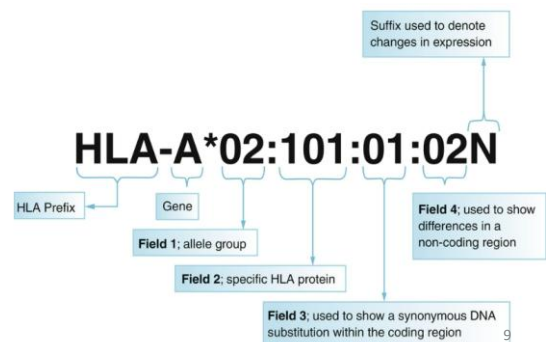
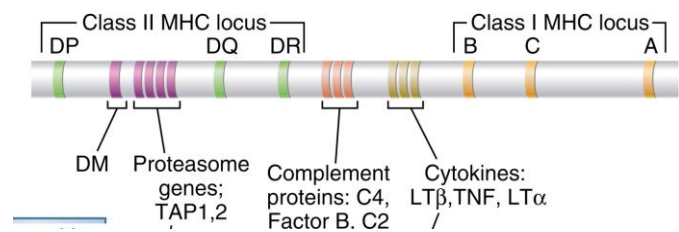
Histocompatibility refers to having similar/enough similarity in histocompatibility antigens between the donor and recipient to facilitate successful transplantations.

Histocompatibility is:

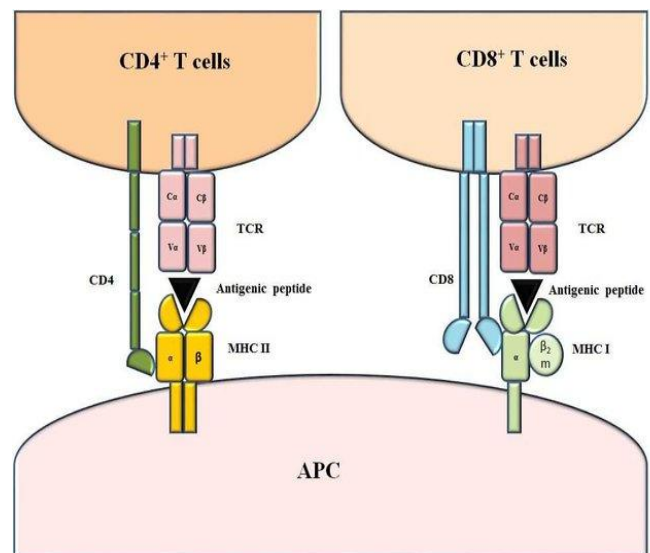
- mainly affected by the **human leukocyte antigen (HLA; aka MHC)**
- To some extent by **minor histocompatibility antigens**. These are small peptides found on the cell surface in association with class I or class II major histocompatibility complex (MHC) molecules.
 - Function: not completely understood. Responsible for fine tuning of successful transplantation. Role in graft vs host diseases in HSCs transplant.

Major Histocompatibility Complex

- MHC genes are highly variable (polymorphic or posses many alleles for each gene).
- Located on chromosome 6.
- Has systemic nomenclature to allow the identification and matching of different alleles.

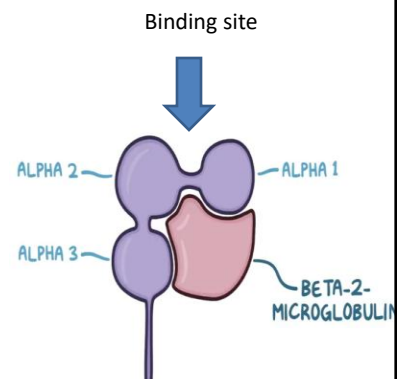


- MHC proteins are synthesized in the endoplasmic reticulum from information encoded in ch6. Then, migrate to the cell membrane.
- Function: display peptide antigens for recognition by T cell receptors.



MHC Class 1 structure

- Three class I MHC genes encode “HLA-A, HLA-B, and HLA-C” class I MHC proteins .
- Each protein pair with β 2-microglobulin= complete class I MHC molecule.
- MHC is polymorphic (1,500 gene isoforms encode the three classes of HLA proteins).
- MHC class I is expressed by all nucleated cells and allows CD8 + T cells to scan the internal environment of cells.

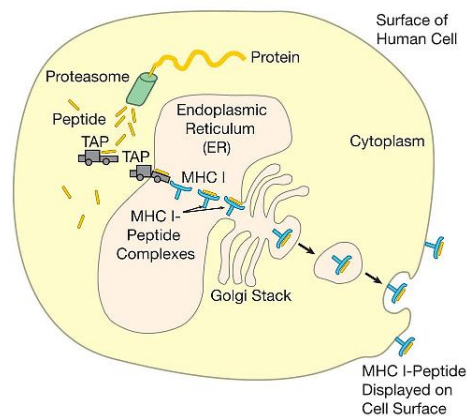


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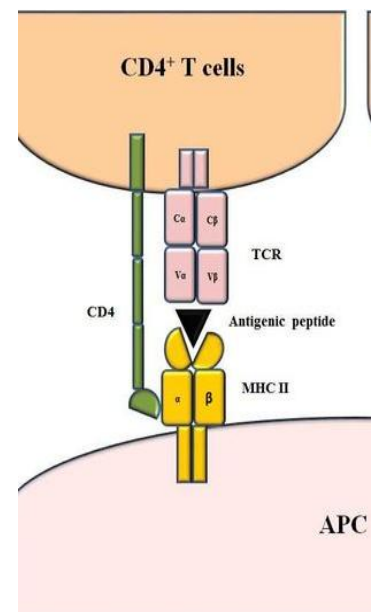
Antigen presentation by MHC class 1

- MHC I display fragments of endogenous proteins (cellular proteins or viral or tumor antigen).
- CTLs inspect the protein fragments displayed by class I MHC molecules.
- Daily turnover to ensure accurate representation of cellular endogenous environment.



MHC class II

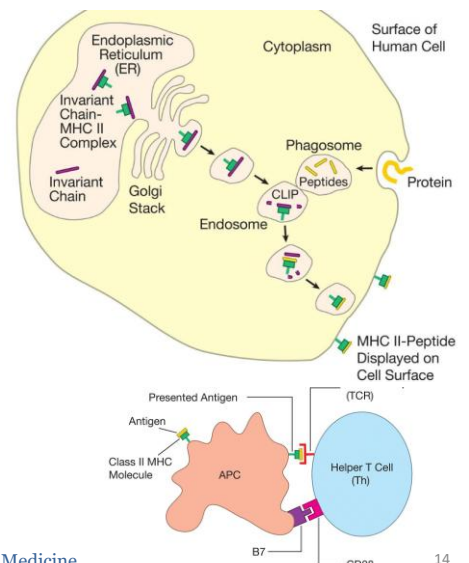
- Class II MHC genes encode HLA-DP, HLA-DQ and HLA-DR proteins.
- Made of an α and a β polypeptide chain.
- Polymorphic population (>700 different versions in humans)
- Expressed on antigen presenting cells, present antigens from outside the cells and are recognized by CD4 T cells.



Why don't MHC II display endogenous proteins?

Due to the invariant chain

- In the ER, MHC II proteins bind to an “invariant chain” to block the binding groove.
- The invariant chain guide MHC II to an endosome that later fuses with a phagosome.
- Then, MHC II is exposed to exogenous antigens.
- MHC expression molecules is increased by cytokines.



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Clinical Significance: MHC and Disease

- Association between MHC gene polymorphism and autoimmune diseases.
- Mutations in different loci were reported in different diseases like type 1 diabetes mellitus (HLA-DR3), Rheumatoid arthritis (HLA-DRB1), multiple sclerosis (HLA-DR2).
- Identifying those gene targets can be beneficial in designing pharmacological targets later one.

Clinical Significance: MHC and transplantation

- All MHC molecules show a high level of allotypic polymorphism (sig.?, exceptions?)
- Thus, MHC molecules play an important role in increasing or decreasing the risk of transplant rejection.
- Matching donors and recipients with highest MHC similarities= favorable success rate.
- The donor's MHC antigens can cause transplant rejection (poorly matched). Donor T and B cells (antibodies) play an important role in graft-versus host disease.

Immune privileged sites

- Immune-privileged sites are organs that are protected, to a variable degree, from the natural immune responses.
- Include brain, eye, testes, placenta, and fetus.
- Therefore, there will be a lack of immune responses to tissue transplanted into any of those sites, HLA matching is not a must (corneal transplant).

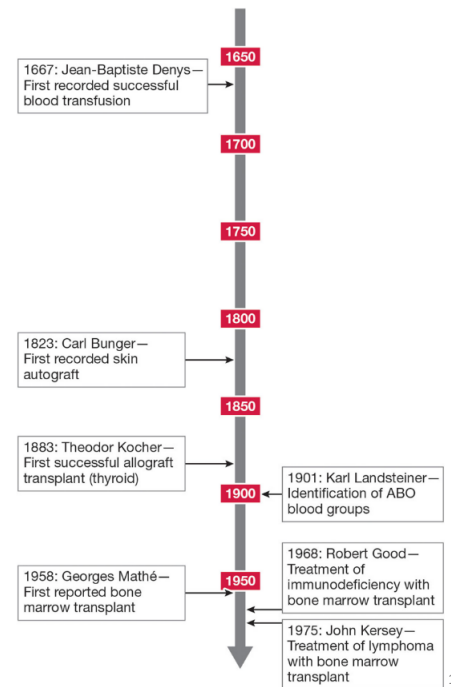
Transplantation & Immune Response

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Why is it needed?

- Burn victims
- Traumatic injury
- Organ Failure
- Tissue reconstruction
- Cancer



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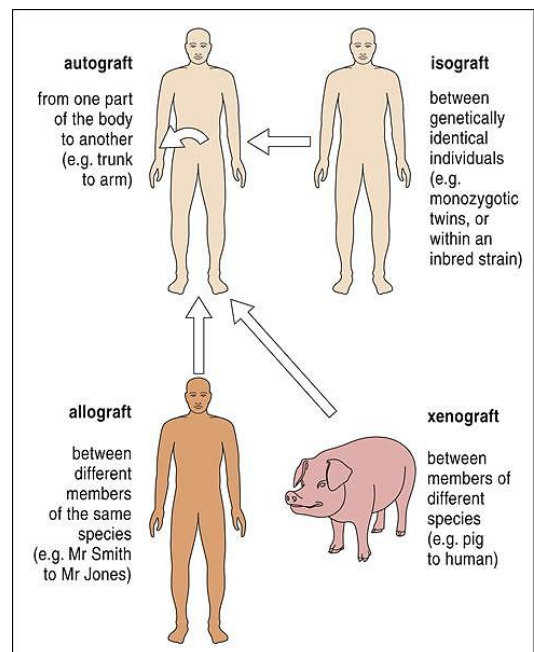
Terms

- **Implant:** Insertion of artificial object in the body
- **Transplant:** the replacement of nonfunctioning organs/tissue with healthy ones from another living organism.
- **Transfusion:** transfer of circulating blood cells or plasma from one individual to another.
- **Grafts:** the cells, tissues or organs harvested from the donor and transplanted into a recipient.
- **Donor:** person/animal donating the tissue or organ.
- **Recipient:** person/animal receiving the tissue or organ.

- **Rejection:** Immune response mediated events leading to the damage of the transplanted organ/tissue.
- **Cadaveric transplant:** use of organ/tissue from diseased donor.
- **Orthotopic transplantation:** If the graft is placed into its normal anatomic location.
- **Heterotopic transplantation:** If the graft is placed in a different site (other than its anatomical location).
- **Alloantigens** are foreign antigens in the graft [transplanted between two genetically different individuals (allografts)].
- **Xenoantigens:** Foreign antigens in a xenograft.

Types of grafts

- 1- Autologous graft (autograft).
- 2- Syngeneic graft (isograft).
- 3- Allogeneic graft (allograft).
- 4- Xenogeneic graft (xenograft)



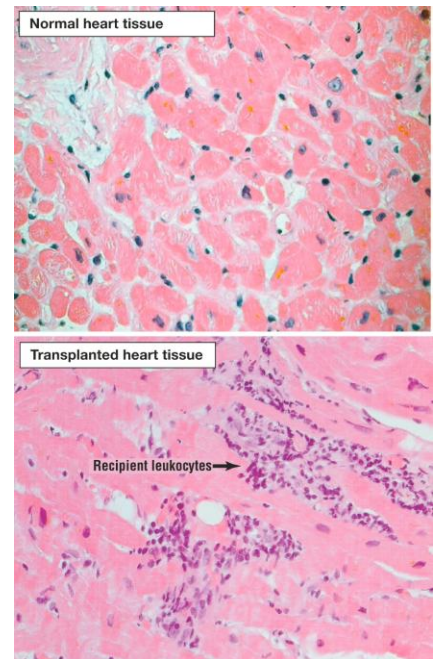
Brostoff J, Male D, et al: Immunology, ed 7, Philadelphia, 2006, Mosby.

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Transplant outcome

- The immune system may treat a transplant as a foreign entity.
- Proper selection of matched graft and suppression of host immune response are imp. for optimal outcome.
- Interruption of blood supply during surgery can cause ischemic damage (ischemia reperfusion injury), leading to expression of DAMPs on graft.
- Transplant rejection is attributed to innate and adaptive immune response.

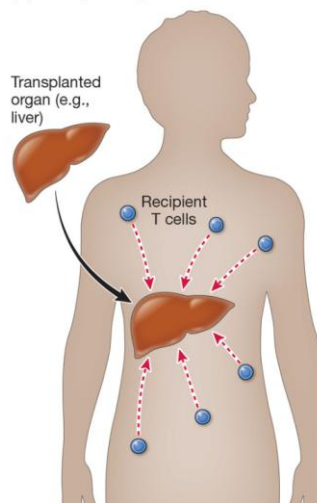


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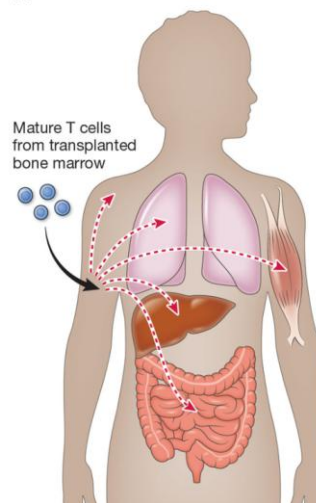
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Outcome: Alloreaactions for transplanted tissue

(A) Transplant rejection



(B) Graft-versus-host disease



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What causes rejection?

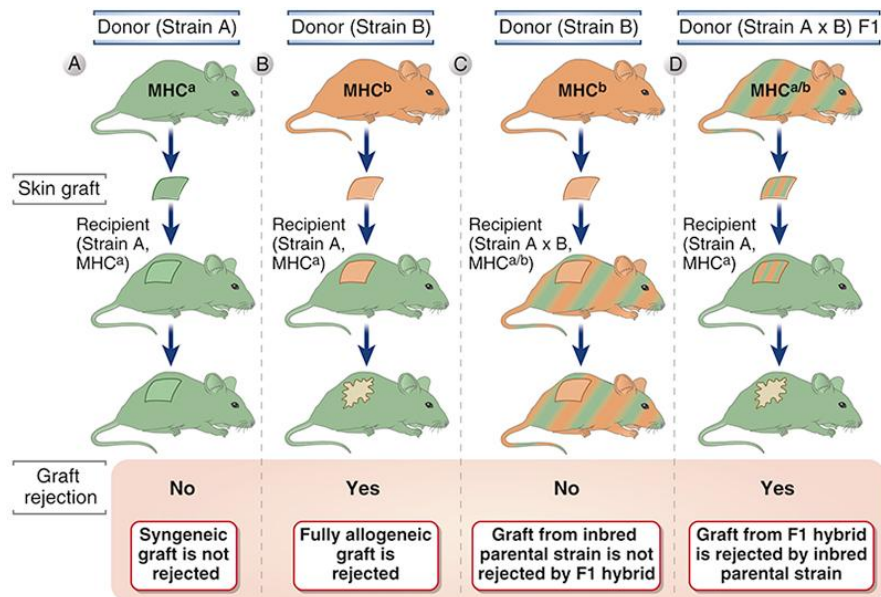
Rejection reactions are attributed to the major histocompatibility complex (MHC) and other polymorphic antigens.

1- **Rapid rejection reactions**, attributed to MHC molecules because they bind and present peptides to T cells.

2- **Slower rejection reaction**, attributed to minor histocompatibility antigens.

Allorecognition refers to recipient immune responses mounted against MHC expressed in an allograft (i.e, coming from the donor).

Genetics of graft rejection



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For Illustration

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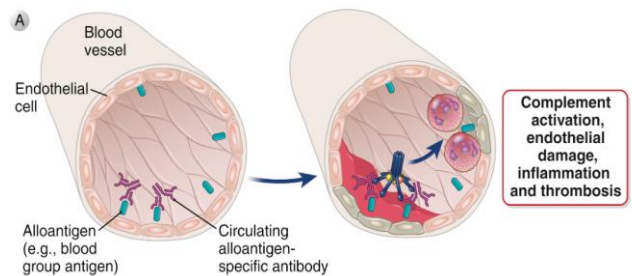
Pattern of Allograft Rejection

1- Hyperacute rejection

Rare occurrence. Characterized by thrombotic occlusion of the graft vasculature that begins within minutes to hours after host and graft blood vessels are joined.

Hyperacute rejection can be mediated by:

- 1- preexisting IgM alloantibodies against the ABO antigens
- 2- IgG developed by previous exposure to alloantigen (previous transplant or transfusions).
- 3- Natural antibodies against different species transplant (xenograft or xenotransplantation).



2- Acute rejection:

Injury to the graft parenchyma and blood vessels mediated by alloreactive T cells and antibodies. Acute rejection may occur following the reduction of the immunosuppression protocol.

A) Acute cellular rejection: CTL-mediated killing of graft parenchymal cells and endothelial cells. Helper T cells induce inflammation through the release of cytokines.

B) Acute humoral rejection: Binding of the alloantibodies (against MHC molecules) to the endothelial cells cause endothelial injury and intravascular thrombosis that result in graft destruction.

3- Chronic rejection

Major cause of transplant failure. Develops during months or years.

Mechanism: related to activation of alloreactive T cells, secretion of inflammatory cytokines (IFN- γ). Continuous inflammation can lead to development of fibrous tissue and fibrosis.

Pathological picture depend on the organ. For example,

- i) Kidney and heart show vascular occlusion and interstitial fibrosis, leading to organ failure.
- ii) Liver transplants show fibrotic and nonfunctional bile ducts.

Mechanisms Driving Solid Organ Transplant Alloreaactions

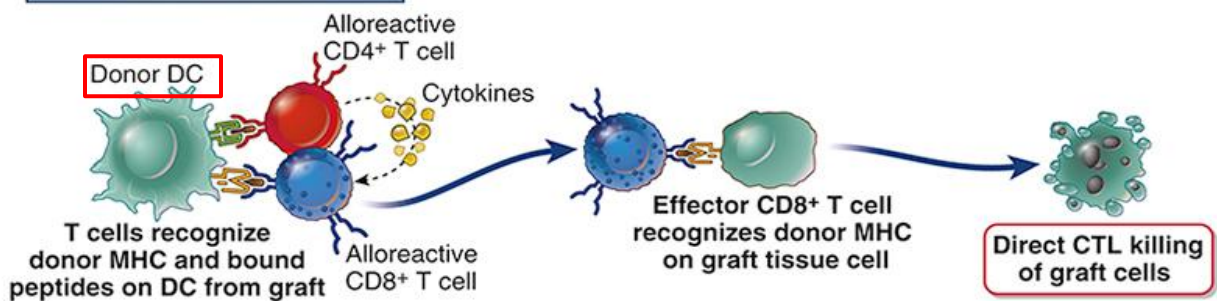
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Recognition of alloantigen by T cells “direct recognition”

CD8⁺ mediated cell death

A Direct alloantigen recognition

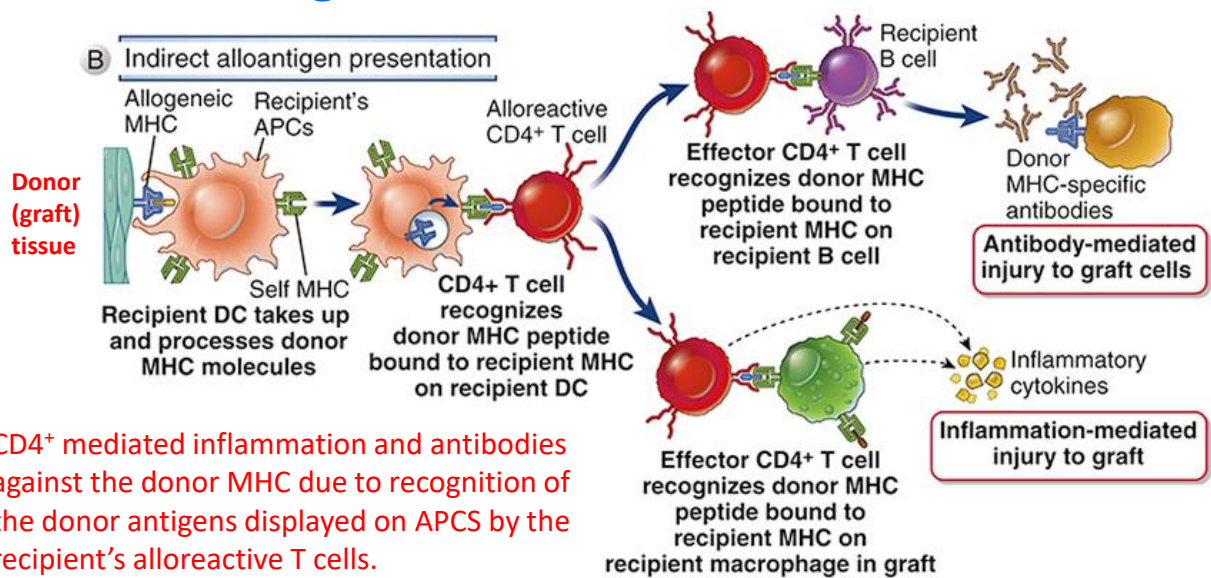


Recognition of intact, unprocessed MHC molecules on grafts by the recipient's T cells. Recognition promotes the differentiation into effector T cells which will specifically target and kill the graft cells.

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Indirect recognition

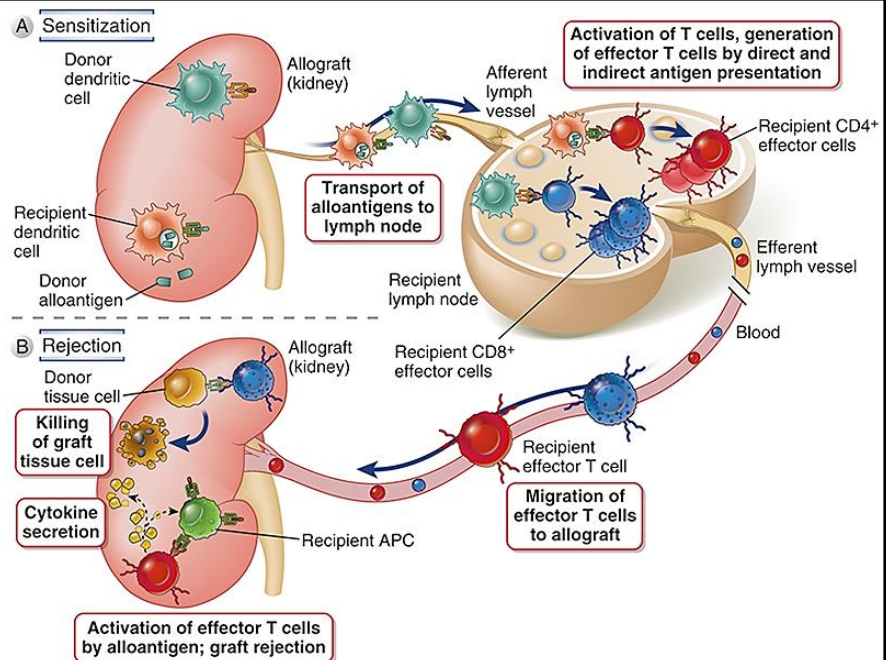


CD4⁺ mediated inflammation and antibodies against the donor MHC due to recognition of the donor antigens displayed on APCs by the recipient's alloreactive T cells.

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Immune cycle leading to graft rejection



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Chain of events leading to graft rejection:

- 1- T cell responses to an organ graft are initiated in the draining lymph nodes. Most organs contain resident antigen presenting cells “dendritic cells”. Upon transplanting these organs, the allogenic recipient is receiving the donated organ as well as the donor’s antigen presenting cells that express the donor’s major histocompatibility complex.
- 2- Those donor antigen presenting cells can migrate to the draining lymph node and present, on their surface, unprocessed allogeneic class I or class II MHC molecules to the recipient’s CD 8+ and CD 4+ T cells, respectively (direct MHC allorecognition).
- 3- Simultaneously, the recipient dendritic cells also migrate into the graft, pick up the graft’s alloantigens, transport them back to the draining lymph nodes. Next, the helper T-cell response and B cell response are activated (this is the indirect pathway).
- 4- Naive helper- and cytotoxic-T lymphocytes that are normally found in the lymph node encounter these alloantigens. This induces their proliferation and differentiation into effector helper T cells and cytotoxic T lymphocytes. The effector cells migrate back into the graft and mediate rejection. Cytotoxic T cells activated by the direct mechanism will induce cell death or apoptosis in the graft cells. Helper T cells will induce damage through the secretion of inflammatory cytokines.
- 5- In addition, antibodies against the graft antigens (also named donor-specific antibodies) also contribute to the rejection events. Helper T-cells as well as the allogenic major histocompatibility complex molecules induce the activation of B cells. Antibodies mediate their function via opsonization and complement activation.

GRAFT-VERSUS-HOST DISEASE

SECTION of TRANSPLANTED
or DONATED TISSUE (e.g. bone
marrow or peripheral blood)



TISSUES of the PERSON
RECEIVING the TRANSPLANT

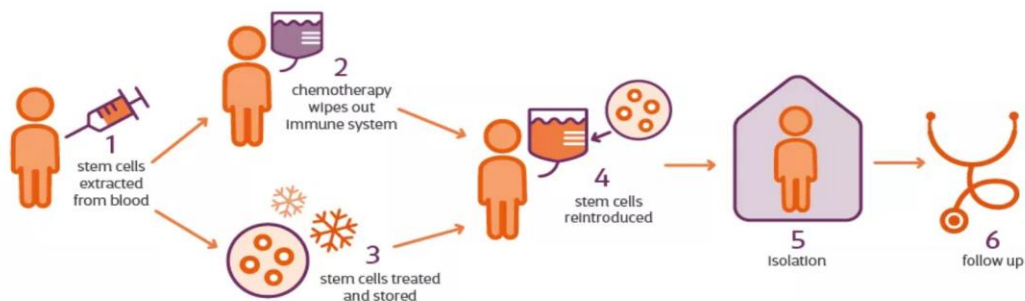


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Stem Cell Transplant

- Hematopoietic stem cell transplant has been used to treat certain hematological cancers (like leukemia and lymphoma), treat several immune disorders & blood disorders like sickle cell disease and thalassemia
- Progenitor stem cells are extracted from the peripheral blood, cord blood or bone marrow.



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- In the GVHD, the graft tissue (donor) attacks the graft recipient (host). This immune disease results from the reaction of grafted mature T cells of the donor with alloantigens of the host (recipient).
- HLA matching and the degree of immunologic difference between the graft and the host play critical role in disease development.
- The number of immune cells in the donor graft plays a major role in disease development.
 - Therefore, graft versus host disease is more frequently encountered in hematopoietic stem cell transplant (HSCT) >> solid organ transplant.
 - Solid organs with higher number of T cells (liver, lung and S. intestine) has higher risk than other organs.

GVHD risk factors in HSCT

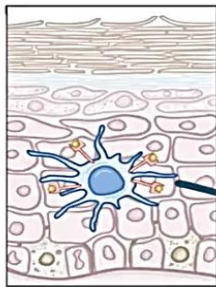
- 1- HLA incompatibility (mainly minor histocompatibility antigens).
- 2- Patient age
- 3- Female gender (& multiple pregnancy)
- 4- Aggressive host immunosuppression (initiation). Rapid tapering of immunosuppression medications.
- 5- Stem cell source (peripheral blood > bone marrow > cord blood).

Mechanism of Graft vs Host disease

CD4 mediated

HOST'S DENDRITIC CELLS

1



Host/recipient's

DC

- VARIOUS HOST TISSUES (e.g. lining of the skin)
- PICK UP ANTIGENS & TRAVEL to LYMPH NODE
- MEET HELPER T-CELLS
- PRESENT ANTIGENS upon their MHC CLASS II MOLECULES



interleukin-2

interferon-gamma

ACTIVATES PHAGOCYTES (e.g. macrophages)



* lysosomal enzymes

HELP T CELLS PROLIFERATE
INDUCE EXPRESSION of more
MHC CLASS II MOLECULES on
HOST'S CELLS

EASIER for DONOR'S HELPER
T-CELLS & HOST'S ANTIGEN-
PRESENTING CELLS to INTERACT

RELEASE PRO-INFLAMMATORY
CYTOKINES (e.g. tumor necrosis
factor-alpha, interleukin-1, and
interleukin-6)

RECRUITS NEW IMMUNE CELLS to AREA

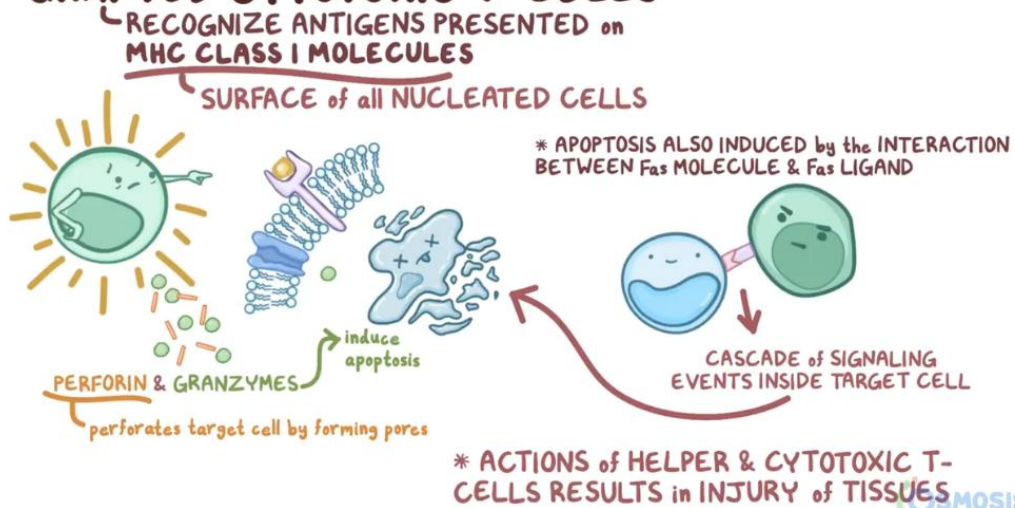
Graft/donor T-cell

TYPE 1 HELPER
T-CELL (Th1)

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GVHD (CD8-mediated)

GRAFTED CYTOTOXIC T-CELLS



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Development of graft vs host disease mechanism:

- 1- Donor's T cells recognizing a recipient's antigen presented on dendritic cells. This will induce the activation and proliferation of the donor's (graft) T-helper cells.
- 2- Activated T-helper cells will release cytokines (TNF alpha, IL-2, IFN gamma), activate phagocytes and induce inflammatory reactions as well as cell apoptosis in the recipient organs. Simultaneously, donor's cytotoxic T cells are activated upon encountering recipient cells expressing the major histocompatibility complex class I. They mediate cell destruction by FAS and perforin/granzymes mediated cytotoxicity.

Effector phase: Expanded cytotoxic T cells, natural killer cells, Th cells and inflammatory cytokines cause direct tissue destruction. Innate immune cells (neutrophils, monocytes) also significantly contribute. If the disease progresses, B cells also become involved.

Classification

Acute GVHD

- If develop in the first 3 months following transplantation.
 - Skin manifestations include rash, erythema of the palms and soles. Severe cases can present with full-thickness skin loss (necrolysis).

Signs of Acute GVHD

Skin

- Very faint to severe sunburn-like rashes
- Blisters

Stomach

- Nausea that doesn't go away
- Loss of appetite
- Vomiting (throwing up)
- Feeling full after eating very little

Intestines

- Diarrhea
- Belly pain that does not go away
- Feeling bloated, or full of gas
- Blood in your stool

Liver

- Jaundice (your skin or eyes look yellow)
- Dark (tea-colored) urine
- Pain in the upper part of your belly
- Swelling in your legs or belly

Chronic GVHD

symptoms develop > 3 months after transplantation. Multi-organs are involved (skin, joints, eyes, lungs, GI and oral).

- Oral cavity- burning sensation, pain, ulceration, white lace-like discoloration (lichenoid appearance).
- Eyes-burning and photophobia.
- Lungs- shortness of breath and coughing
- Liver-jaundiced
- Muscles-cramping and weakness.
- Kidney-nephrotic syndrome

Signs of Chronic GVHD

>3 months after grafting

Skin and nails

- Skin texture changes (thickening)
- Nail changes
- Rash
- Unusual hair loss or thinning
- Itchy skin

Joints and muscles

- Arthritis-like symptoms (pain and stiffness)
- Muscle pain, cramps or weakness

Eyes

- Dry eyes
- Irritation that doesn't go away
- Blurred vision
- Teary eyes

Mouth

- Trouble opening your mouth
- Sores
- Irritation that doesn't go away
- Chapped lips
- Pain

Lungs

- Cough that doesn't go away
- Shortness of breath
- Trouble breathing

Digestive system

- Nausea or vomiting (throwing up)
- Diarrhea
- Stomach pain or cramping

Chronic GVHD in Oral cavity

Prevalence: around 45-83 % of chronic GVHD patients.

Symptoms:

Trismus

Pain/burning sensation

Ulceration

Xerostomia

Oral mucositis

white lace-like discoloration (lichenoid lesions).

Treatment:

Corticosteroids with or without immunosuppressants.

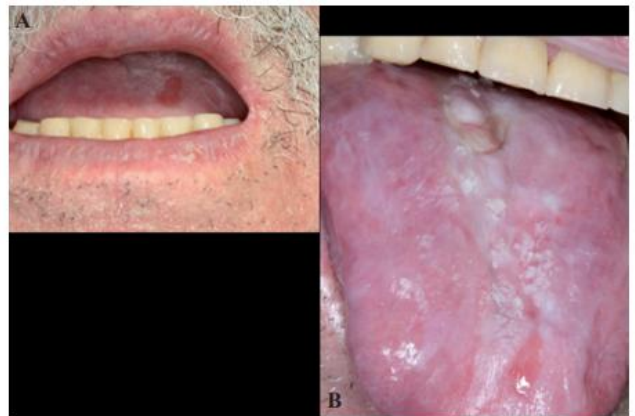


Fig. 2. Clinical features of oral cGVHD. Mouth opening restriction due to sclerosis (A) and lichenoid lesions on tongue (B).

[Graft-versus-host disease affecting oral cavity. A review - PMC \(nih.gov\)](#)

Strategies to increase transplant success

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Laboratory tests

1- Blood group matching.

The blood group of the donor and recipient should be compatible in solid organ transplants

2- Cross matching.

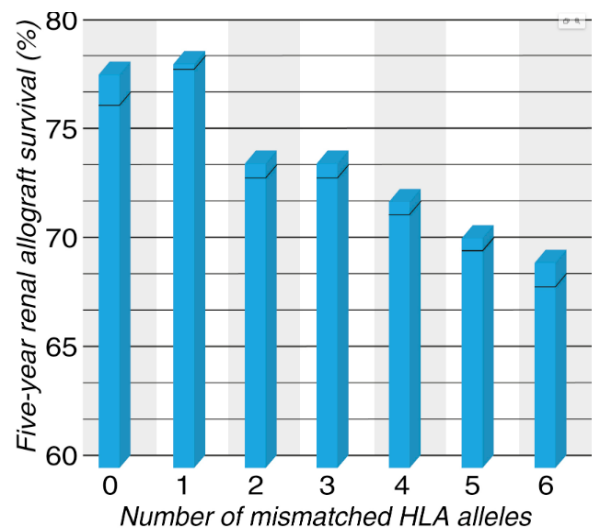
Screens for cytotoxic antibodies in the recipient's serum against the graft tissue. It detects the lysis of donor lymphocytes by the recipient's serum.

3- Tissue or HLA typing

Checks the similarity in the MHC genes between the donor and the recipient. Genetic material from the donor and recipient are compared by DNA sequencing.

Impact of HLA matching on graft survival

- MHC matching between the donor and recipient significantly increases the chance of renal allograft survival.
- The data shown are for deceased donor (cadaver) grafts.
- Is MHC matching a must? Do we have exceptions?

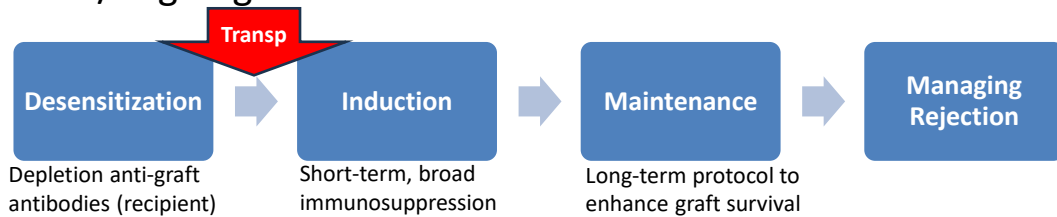


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Preventing Rejection

- Immunosuppressive cocktails:
 - Anti-proliferatives.
 - Inhibitors of T-cell activation.
 - Anti-inflammatory agents.
 - Biologics (like antibodies).
- Future/ongoing Research: Induction of tolerance



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Examples

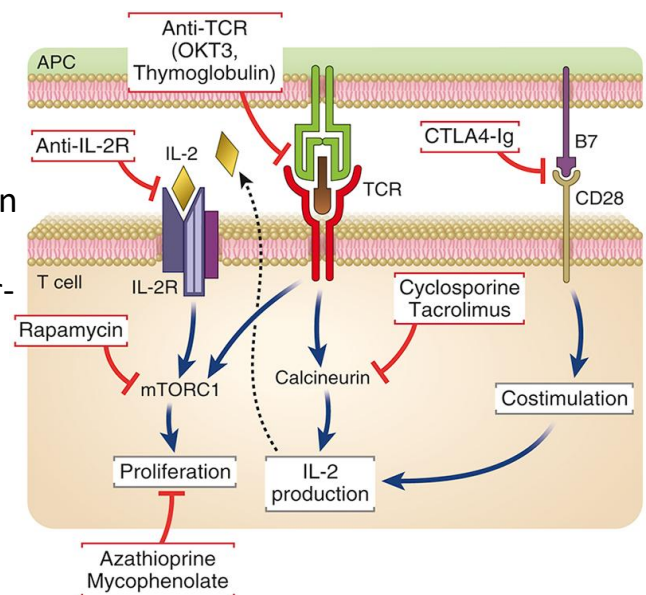
1. Inhibitors of T-cell signaling pathways:

a) **Cyclosporine**, inhibits the transcription of genes encoding IL-2.

b) **Rapamycin**, inhibits the growth factor-mediated T cell proliferation.

c) **Monoclonal antibodies**.

Anti-CD25 antibody inhibit T-cell activation via blocking the interleukin-2 receptor.



2. Anti-inflammatory agents, like corticosteroids. Effect of use can include:

- inhibition of pro-inflammatory cytokine secretion (IL-1, IL-3, IL-4, IL-5, IL-8 (CXCL8), $\text{TNF}\alpha$).
- Inhibition of nitric oxide synthase
- Inhibition of adhesion molecule expression (reduced leukocyte migration).
- Induction of endonucleases (causing apoptosis of lymphocytes and eosinophils).

Dental Implications

A) Medication-induced Immunosuppression:

- Risk for opportunistic infection.
- Risk of infection complication
- Drug-induced gingival hyperplasia.
- Delayed wound healing.

B) Oral health implication in GVHD

Xenograft transplantation

- Fueled by shortage of available human organs. To help prevent transplant tourism.
- Pigs have physiologic similarities to humans, and some breeds have organs with appropriate size for human recipients.
- Pig skin was used for temporary coverage of damaged areas in human burn victims. Pig valves were used for heart valve replacement.
- Limitations: possibility of zoonotic infections through xenotransplantation, public acceptance and non-human epitopes (removed by genetic engineering).

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Results of Two Cases of Pig-to-Human Kidney Xenotransplantation

Genetically modified kidney xenografts from pigs remained viable and functioning in brain-dead human recipients for 54 hours, without signs of hyperacute rejection.

> [Circulation](#). 2022 Jun 21;145(25):1870-1871. doi: 10.1161/CIRCULATIONAHA.122.060418. Epub 2022 Jun 21.

First Pig-to-Human Heart Transplant Marks a Milestone in Xenotransplantation

Bridget M Kuehn

Genetically modified pig heart was successfully transplanted into a 57-year-old man in the end stage of heart disease. The patient died after 2 months

[Results of Two Cases of Pig-to-Human Kidney Xenotransplantation | NEJM](#)
[After the First Pig-to-Human Heart Transplant, Scientists Look to the Future of Cardiac Xenotransplantation | Cardiology | JAMA | JAMA Network](#)

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Big picture

- Terms and definitions ★
- Direct vs indirect recognition
- Different stages of transplant rejection
- GVHD, mechanism as well as symptoms of acute vs chronic. ★

References

